Asymptotic Saturation of Projective Resolution: Spectral Growth along the Projective Cascade

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Abstract

We investigate the spectral mechanism governing mode stabilisation in the Cosmochrony projective cascade framework. Starting from the admissibility condition defining the projective resolution Λ_{proj}, we derive its dependence on the isoperimetric capacity of the cascade graph and show that $\Lambda_{\text{proj}} \asymp h(G)^2$. For expander families satisfying spectral-isoperimetric saturation, this implies $\Lambda_{\text{proj}} \asymp \lambda_2$.

In the Lubotzky–Phillips–Sarnak (LPS) graph model with fixed prime p, the spectral gap converges to a constant, making Λ_{proj} asymptotically static. In this regime, the stabilisation of modes is governed not by the admissibility threshold but by saturation of the cumulative spectral count below Λ_{proj}.

Combining this counting mechanism with the representation structure of ADE substrates yields a factorised prediction for mass ratios,

$$\frac{\mathcal{M}_i}{\mathcal{M}_j} \propto \frac{F_{\text{KM}}(\lambda_i)}{F_{\text{KM}}(\lambda_j)} \cdot \frac{\dim \rho_{\lambda_j}}{\dim \rho_{\lambda_i}},$$

where F_{KM} is the Kesten–McKay cumulative distribution function.

Numerical evaluation shows that this single-level mechanism produces mass ratios of order unity and inverts the ordering expected from the admissibility envelope. We identify the absence of a dynamical threshold $\Lambda_{\text{proj}}(n)$ as the origin of both limitations and outline directions for restoring threshold dynamics within the Cosmochrony framework.

Keywords: Spectral graph theory; Ramanujan graphs; Expander graphs; Cheeger inequality; Kesten–McKay distribution; Graph Laplacian spectrum; Emergent mass hierarchy

Contents

1 Introduction

1.1 The stratigraphic problem

The Bounded Admissibility framework [2] associates to each cascade rank n an energy scale $E_n = E_{\mathrm{P}}/n$ and a spectral admissibility condition: a mode of eigenvalue λ is projectable at rank n if and only if $\lambda \leq \Lambda_{\mathrm{proj}}(n)$, where $\Lambda_{\mathrm{proj}}(n)$ is the projective resolution at that stage of the cascade. The *stabilisation rank* of a mode is the first n at which it ceases to be admissible, and the corresponding stabilisation energy is taken as a proxy for the physical mass of the associated particle sector.

For this picture to produce a viable mass hierarchy, three conditions must hold simultaneously. First, the projective resolution must decrease with rank, so that earlier stabilisations correspond to higher masses. Second, the set of stabilisation ranks must be discrete, so that the spectrum of effective masses is not a continuum. Third, the inter-level separations must be large enough to account for the observed mass ratios between generations, which span five to seventeen orders of magnitude from the electron to the top quark.

The first two conditions were addressed in [5]. The third – which requires a quantitative law for $\Lambda_{\mathrm{proj}}(n)$ – was left as an open problem there and is the subject of the present paper.

1.2 What SpectralStratigraphy established

The companion paper [5] established two results that together constrain the form of $\Lambda_{\mathrm{proj}}(n)$ from above and below.

No-go for scalar calibration (Proposition 2 of [5]). A threshold of the form $E^*(\lambda) = E_0\lambda^\beta$ cannot produce a discrete stratigraphic profile for any choice of parameters (α, β). This rules out any explanation of the discreteness of the generation spectrum by rescaling or re-calibrating the energy-eigenvalue relation. The discretisation must therefore originate in the internal structure of the projective substrate.

Three-level structure from ADE binary groups (Proposition 3 of [5]). For four specific pairs (G, S) drawn from the binary tetrahedral, octahedral, and icosahedral groups – $2T$ with order-3 generators, $2O$ with order-3 generators, and $2I$ with order-4 or order-5 generators – the Cayley graph Laplacian has exactly three distinct non-zero eigenvalues. Each eigenvalue level corresponds to a block of irreducible representations. Consequently, the stratigraphy defined by the Born–Infeld admissibility envelope $A^{\max}(\lambda) \propto 1/\sqrt{\lambda}$ produces exactly three stratigraphic classes, ordered by decreasing admissibility window.

These two results together establish the following separation of roles, which is the central logical outcome of [5]:

> *Group topology determines the number of stratigraphic levels. Projective dynamics determines the inter-level separations.*

1.3 What remained open

Despite the clean three-level structure, the numerical inter-level separations produced by the ADE examples are small: between 0.08 and 0.18 log-decades in cascade depth $\log n$, as reported in Table 2 of [5]. This is far too small to account for mass ratios of order 10^5–10^{17}.

The gap is not a defect of the group-theoretic mechanism. It reflects the fact that the spectral ratios λ_ℓ/λ_h for the ADE Cayley graphs lie in the range $[1.5, 2.0]$: the eigenvalue levels are spectrally close. Amplifying these small spectral differences into large mass ratios requires a rapidly growing function $\Lambda_{\mathrm{proj}}(n)$.

The missing ingredient is therefore precisely the *growth law* of $\Lambda_{\mathrm{proj}}(n)$ along the cascade trajectory. This law must be derived from within the framework, not imposed as an external ansatz: the no-go result rules out scalar power-law calibrations, and any other ad hoc choice would undermine the predictive content of the stratigraphy.

1.4 Aim and approach of the present paper

This paper proposes a derivation of $\Lambda_{\mathrm{proj}}(n)$ grounded in the global coherence properties of the cascade graph $G(n)$. The growth of $G(n)$ with n — in the sense that $|V(G(n))| \sim n$ — encodes the progressive unfolding of the observable algebra along the projective cascade: at each cascade rank the effective graph $G(n)$ represents the relational structure accessible to the projection Π at that depth, and its isoperimetric capacity controls which spectral modes of the substrate are projectively admissible.

The key observation is that $\Lambda_{\mathrm{proj}}(n)$ is entirely controlled by the global projective flux bound $c_\chi(n)$:

$$\Lambda_{\mathrm{proj}}(n) = \frac{c_\chi(n)^2}{A_{\mathrm{min}}^2}.$$

The dynamical question reduces to: *how does $c_\chi(n)$ depend on the cascade rank?* We argue that $c_\chi(n)$ is bounded by the isoperimetric capacity of $G(n)$, measured by the Cheeger constant $h(G(n))$, which is in turn controlled by the algebraic connectivity $\lambda_2(n)$ via the discrete Cheeger inequality.

The logical chain of the derivation is:

$$\text{global coherence} \;\longrightarrow\; c_\chi(n) \;\longrightarrow\; h\big(G(n)\big) \;\longrightarrow\; \lambda_2(n) \;\longrightarrow\; \Lambda_{\mathrm{proj}}(n).$$

Two structural hypotheses are required: a *projective coherence closure* (Hypothesis 2.1) and a *spectral-isoperimetric saturation* condition (Hypothesis 2.2). Both are stated explicitly and treated as independent structural postulates.

The main results of the paper are as follows.

- **Proposition 2.1**: $\Lambda_{\mathrm{proj}}(n) \asymp h(G(n))^2$ under Hypothesis 2.1 alone.
- **Corollary 2.1**: $\lambda_2(n)^2 \lesssim \Lambda_{\mathrm{proj}}(n) \lesssim \lambda_2(n)$ via the Cheeger inequality, with no additional hypotheses.
- **Corollary 2.2**: $\Lambda_{\mathrm{proj}}(n) \asymp \lambda_2(n)$ under both hypotheses.

The programmatic consequence is that the problem of inter-generation mass ratios reduces to the problem of computing $\lambda_2(n)$ along the cascade trajectory.

1.5 Structure of the paper

Section 2 establishes the formal framework. Section 3 introduces the LPS graph model and identifies the cascade index. Section 4 analyses the spectral behaviour of LPS graphs and derives the growth law for $N(\lambda; n)$. Section 5 evaluates the predicted mass ratios numerically. Section 6 discusses limitations and open directions. Section 7 concludes.

1.6 Position within the Cosmochrony programme

This paper explores a specific model of the projective cascade — LPS Ramanujan graphs with fixed prime p — and establishes a structural negative result: the admissibility threshold Λ_{proj} is asymptotically static in this model, which is insufficient to produce the observed inter-generation mass hierarchy.

The derivation of the mass hierarchy within the Cosmochrony programme rests on a distinct mechanism established in the O-series: the transfer chain $c_\chi \to \delta_{\mathrm{pair}} \to \beta^*$, proved unconditionally in [6] and validated numerically across all conjugate pairs in [4]. That chain operates within a fixed Heisenberg Cayley graph $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ via the decay of the pair capacity $\sigma_{\mathrm{pair}}(n)$ along the BFS cascade, and does not depend on the LPS model studied here.

The present paper contributes three elements to the broader programme: the isoperimetric law $\Lambda_{\mathrm{proj}} \asymp h(G)^2$ (Proposition 2.1), the spectral analysis of LPS graphs via the Kesten–McKay distribution, and the identification of direction (O1) — allowing $p(n) \to \infty$ jointly with n — as the natural candidate for a dynamical cosmological threshold compatible with the observed mass ordering. The development of that cosmological model is left to future work.

2 Formal Framework

2.1 Notation and basic objects

Throughout this paper, $n \in \mathbb{N}_{>0}$ denotes the *cascade rank*, an integer-valued depth parameter indexing successive stages of the projective cascade. It is distinct from the cardinality of any group or graph. The energy available at rank n is

$$E_n \;=\; \frac{E_{\mathrm{P}}}{n}, \tag{1}$$

where E_{P} is the Planck energy, in agreement with the cascade scaling of [5].

Let $G(n)$ denote the *effective cascade graph* at rank n: a connected, $d(n)$-regular, undirected graph encoding the relational substrate at stage n. The vertex set is $V(n)$, the edge set $E(n)$, and the normalised graph Laplacian is $L_{G(n)}$, with eigenvalues $0 = \mu_0 < \mu_1 \le \mu_2 \le \cdots$. The *algebraic connectivity* (Fiedler value) of $G(n)$ is

$$\lambda_2(n) \;:=\; \mu_1\big(L_{G(n)}\big) \;>\; 0. \tag{2}$$

The *Cheeger constant* of $G(n)$ is

$$h\big(G(n)\big) \;:=\; \min_{\substack{S\subset V(n)\\ 0<|S|\le|V(n)|/2}} \frac{|\partial S|}{|S|}, \tag{3}$$

where ∂S is the set of edges with exactly one endpoint in S.

2.2 Projective admissibility and flux bound

Following [3, 5], a spectral mode of eigenvalue λ is *projectively admissible* at rank n if the maximal effective amplitude supported by the projective flux bound $c_\chi(n) > 0$ satisfies

$$A^{\max}(\lambda, n) \;:=\; \frac{c_\chi(n)}{\sqrt{\lambda}} \;\ge\; A_{\min}, \tag{4}$$

where $A_{\min} > 0$ is a fixed scale independent of n. The *projective resolution* at rank n is

$$\Lambda_{\mathrm{proj}}(n) \;:=\; \sup\bigl\{\lambda > 0 : A^{\max}(\lambda, n) \geq A_{\min}\bigr\} \;=\; \frac{c_\chi(n)^2}{A_{\min}^2}. \tag{5}$$

All dependence of $\Lambda_{\mathrm{proj}}(n)$ on the cascade rank is carried entirely by $c_\chi(n)$.

2.3 Structural hypotheses

Hypothesis 2.1 (Projective coherence closure)**.** The global projective flux bound satisfies

$$c_\chi(n) \;\asymp\; h\bigl(G(n)\bigr), \tag{6}$$

with positive constants independent of n.

Hypothesis 2.1 is a closure assumption: it identifies the projective flux capacity with the isoperimetric capacity of the cascade graph. It is not derived from the internal dynamics of $G(n)$ and must be treated as an independent structural postulate.

Hypothesis 2.2 (Spectral-isoperimetric saturation)**.** Along the cascade trajectory,

$$h\bigl(G(n)\bigr)^2 \;\asymp\; \lambda_2(n), \tag{7}$$

with constants independent of n.

Hypothesis 2.2 holds when $G(n)$ belongs to a quasi-expander family in which $h \asymp \sqrt{\lambda_2}$ up to bounded factors. It is logically independent of Hypothesis 2.1.

2.4 Main results

Proposition 2.1 (Projective resolution from isoperimetric capacity)**.** *Under Hypothesis 2.1,*

$$\Lambda_{\mathrm{proj}}(n) \;\asymp\; h\bigl(G(n)\bigr)^2. \tag{8}$$

Proof. Immediate from (5) and (6): $\Lambda_{\mathrm{proj}}(n) = c_\chi(n)^2/A_{\min}^2 \asymp h(G(n))^2/A_{\min}^2$. □

Corollary 2.1 (Cheeger enclosure for the projective resolution)**.** *For a $d(n)$-regular graph, the discrete Cheeger inequality [7]*

$$\frac{\lambda_2(n)}{2} \;\leq\; h\bigl(G(n)\bigr) \;\leq\; \sqrt{2\,\lambda_2(n)} \tag{9}$$

combined with Proposition 2.1 gives

$$\frac{\lambda_2(n)^2}{4\,A_{\min}^2} \;\lesssim\; \Lambda_{\mathrm{proj}}(n) \;\lesssim\; \frac{2\,\lambda_2(n)}{A_{\min}^2}. \tag{10}$$

Remark 2.1. The enclosure (10) is the sharpest consequence of Cheeger's inequality alone. The lower bound is quadratic and the upper bound is linear in $\lambda_2(n)$; no further tightening is possible without additional hypotheses on $G(n)$.

Corollary 2.2 (Linear projective-resolution law)**.** *Under Hypotheses 2.1 and 2.2,*

$$\Lambda_{\mathrm{proj}}(n) \;\asymp\; \lambda_2(n). \tag{11}$$

Proof. Combine Proposition 2.1 with Hypothesis 2.2. □

Its physical content is that the maximal projectable eigenvalue at rank n is controlled, up to bounded factors, by the algebraic connectivity of the effective cascade graph.

2.5 Dynamic extension: spectral counting and stabilisation

Definition 2.1 (Spectral counting function)**.**

$$N(\lambda; n) \;:=\; \big|\{k : \lambda_k\big(G(n)\big) \le \lambda\}\big|, \tag{12}$$

where eigenvalues are counted with multiplicity.

Definition 2.2 (Projectable mode count)**.**

$$N_{\mathrm{proj}}(n) \;:=\; N\big(\Lambda_{\mathrm{proj}};\, n\big). \tag{13}$$

Λ_{proj} fixes the spectral support of projectable modes; $N_{\mathrm{proj}}(n)$ counts how many such modes exist at rank n. These are independent quantities: Λ_{proj} depends on $c_\chi(n)$, while $N_{\mathrm{proj}}(n)$ depends on the spectral density of $G(n)$.

Hypothesis 2.3 (Representational saturation)**.** A spectral mode at eigenvalue level λ_i of the ADE substrate graph stabilises when its cumulative spectral count reaches the dimension of the corresponding irreducible representation block:

$$M(\lambda_i) \;:=\; k\, \dim \rho_{\lambda_i}, \tag{14}$$

where $k \in \mathbb{N}_{>0}$ is a universal constant independent of i.

Definition 2.3 (Stabilisation rank)**.** Under Hypothesis 2.3,

$$n_{\mathrm{proj}}(\lambda_i) \;:=\; \inf\big\{n \in \mathbb{N}_{>0} : \lambda_i \le \Lambda_{\mathrm{proj}} \;\text{ and }\; N(\lambda_i;\, n) \ge k\, \dim \rho_{\lambda_i}\big\}. \tag{15}$$

The effective stabilisation mass is

$$\mathcal{M}_i \;\propto\; \frac{E_{\mathrm{P}}}{n_{\mathrm{proj}}(\lambda_i)}. \tag{16}$$

Remark 2.2 (Separation of roles). The condition $\lambda_i \le \Lambda_{\mathrm{proj}}$ is a support condition governing which modes can stabilise; $N(\lambda_i; n) \ge k\, \dim \rho_{\lambda_i}$ is a saturation condition governing at which rank. When Λ_{proj} is asymptotically constant, the support condition reduces to a fixed constraint and all dynamics arises from the saturation condition.

3 Choice of a Cascade-Graph Model

3.1 Requirements on the cascade graphs

The following properties are necessary for a family $\{G(n)\}_{n\ge1}$ to be consistent with the projective cascade picture.

(R1) Vertex-transitivity. The projection must not privilege any node; $G(n)$ must be vertex-transitive or quasi-homogeneous, so that $\lambda_2(n)$ is a well-defined global quantity.

(R2) Bounded degree. The local relational flux capacity is finite, so $d(n)$ must remain bounded (or grow at most logarithmically) along the cascade.

(R3) Non-vanishing algebraic connectivity. $\lambda_2(n) > 0$ must hold for all n, ensuring a coherent global projection.

(R4) Controlled spectral gap. $\lambda_2(n)$ must admit explicit analytic lower bounds.

(R5) Constructibility as an explicit sequence. The family must admit an explicit indexing compatible with the cascade rank n.

3.2 Candidate families and selection of LPS graphs

Random regular graphs satisfy (R1)–(R4) only in probability and not constructively, failing (R5). Cayley graphs of finite groups satisfy (R1) and (R4) for families with controlled character sums, but generic sequences need not satisfy (R3). Explicit expander families satisfy all five requirements by construction. Among expanders, *near-Ramanujan* families are optimal: the Alon–Boppana theorem [1] bounds $\lambda_2 \leq 1 - 2\sqrt{d-1}/d + o(1)$, and Ramanujan graphs achieve this bound exactly.

The Lubotzky–Phillips–Sarnak (LPS) construction [9] provides the only currently known explicit infinite families of Ramanujan graphs, and is the natural canonical choice satisfying all five requirements.

3.3 LPS graphs

Let p and q be distinct odd primes, both congruent to $1 \pmod 4$. The LPS graph $X^{p,q}$ is a $(p+1)$-regular Cayley graph on $\mathrm{PSL}(2, \mathbb{F}_q)$, with

$$\left|V(X^{p,q})\right| \;=\; \tfrac{1}{2}\, q(q^2-1). \tag{17}$$

The Ramanujan property gives, for the normalised Laplacian,

$$\lambda_2\big(X^{p,q}\big) \;\geq\; 1 - \frac{2\sqrt{p}}{p+1}, \tag{18}$$

uniformly in q. All five requirements (R1)–(R5) are satisfied.

Remark 3.1 (Laplacian normalisation). Throughout this paper, $L_G = I - A/d$ denotes the normalised graph Laplacian, with eigenvalues in $[0, 2]$. The Cheeger inequality (9) and the Ramanujan bound (18) use this convention consistently.

3.4 Identification of the cascade index

The physical identification is

$$n \;\sim\; \big|V\big(G(n)\big)\big|, \tag{19}$$

interpreting the cascade depth as proportional to the number of active relational nodes. For the LPS family at fixed p,

$$n \;\sim\; \tfrac{1}{2}\, q(q^2-1) \;\sim\; q^3/2 \qquad (q \to \infty), \tag{20}$$

so $q \sim (2n)^{1/3}$. Alternative identifications ($n \sim |E(G(n))|$, $n \sim \log |V|$) are discussed in Section 6.

4 Spectral Behaviour of LPS Graphs

4.1 Algebraic connectivity along the cascade trajectory

Fix p and let $q \to \infty$. From (18) and the Alon–Boppana bound [1],

$$\lambda_2(n) \;\longrightarrow\; \lambda_2^*(p) \;:=\; 1 - \frac{2\sqrt{p}}{p+1} \;>\; 0. \tag{21}$$

Proposition 4.1 (Asymptotic constancy of Λ_{proj}). *Under Hypotheses 2.1 and 2.2, for the LPS family at fixed p,*

$$\Lambda_{\mathrm{proj}}(n) \;\longrightarrow\; \Lambda_0(p) \;:=\; \frac{C\,\lambda_2^*(p)}{A_{\min}^2} \qquad \text{as } n \to \infty. \tag{22}$$

Proposition 4.1 shows that the admissibility threshold is asymptotically static in this model. The dynamical variable driving stabilisation is therefore the saturation condition $N(\lambda_i; n) \geq k \dim \rho_{\lambda_i}$, not the variation of Λ_{proj}.

4.2 Kesten–McKay spectral density

As $q \to \infty$ with p fixed, the empirical spectral measure of $L_{X^{p,q}}$ converges weakly to the *Kesten–McKay measure* [10, 8]

$$\mathrm{d}\mu_{\mathrm{KM}}^{(p)}(\lambda) \;=\; \rho_{\mathrm{KM}}^{(p)}(\lambda)\,\mathrm{d}\lambda, \qquad \rho_{\mathrm{KM}}^{(p)}(\lambda) \;=\; \frac{(p+1)\sqrt{4p-(p+1)^2(1-\lambda)^2}}{2\pi\,p\,\big(1-(1-\lambda)^2\big)}, \tag{23}$$

supported on $[\lambda_-, \lambda_+]$ with $\lambda_\pm = 1 \mp 2\sqrt{p}/(p+1)$. The density is symmetric about $\lambda = 1$: $\rho_{\mathrm{KM}}(\lambda) = \rho_{\mathrm{KM}}(2-\lambda)$.

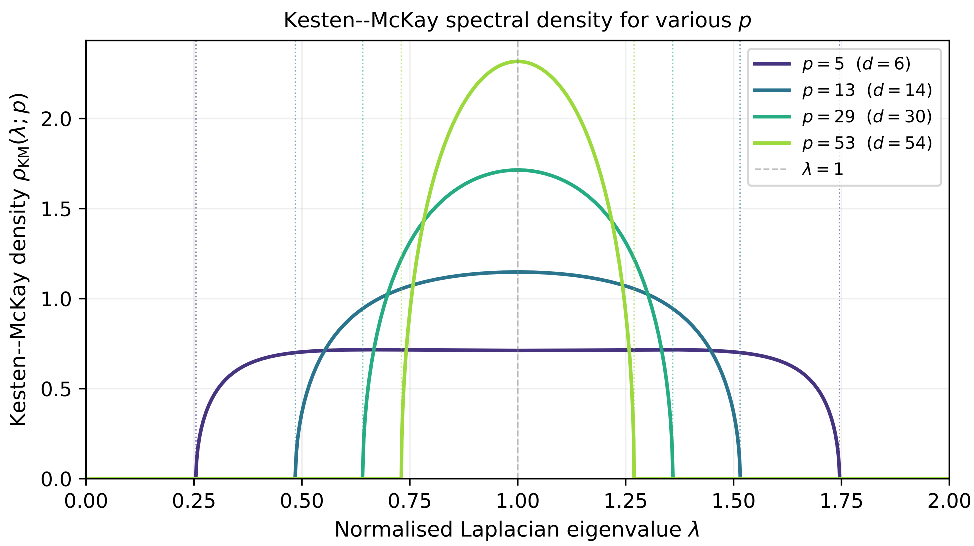

Figure 1: Kesten–McKay spectral density $\rho_{\mathrm{KM}}^{(p)}(\lambda)$ for $(p+1)$-regular graphs with $p \in \{5, 13, 29, 53\}$. The support $[\lambda_-, \lambda_+]$ narrows as p grows. Dashed vertical lines mark the three normalised eigenvalue levels of the $2I$ ord-5 ADE case ($\lambda_1 = 20/24$, $\lambda_2 = 1$, $\lambda_3 = 30/24$). All three levels lie inside the support for $p \leq 53$; the structural identity $F_{\mathrm{KM}}(1) = 1/2$ follows from the symmetry of the density about $\lambda = 1$.

Remark 4.1 (Relation to the S^3 spectral measure). As $p \to \infty$, the Kesten–McKay measure converges to the spectral measure of the S^3 Laplacian [3]. The LPS family therefore interpolates between the discrete Ramanujan regime and the S^3 continuum.

4.3 Monotonicity and linear growth of the spectral counting function

Write $F_{\mathrm{KM}}(\lambda) := \mu_{\mathrm{KM}}^{(p)}((-\infty, \lambda])$ for the cumulative distribution function. By weak convergence and the identification (19):

$$N(\lambda;\, n) \;\sim\; F_{\mathrm{KM}}(\lambda) \cdot n \qquad (n \to \infty). \tag{24}$$

Lemma 4.1 (Monotonicity)**.** *For λ in the interior of $[\lambda_-, \lambda_+]$, $N(\lambda; n)$ is strictly increasing in n for all sufficiently large n.*

Proof. Since $F_{\mathrm{KM}}(\lambda) > 0$ for λ in the interior of the support and $|V(G(n))| \sim n$ is strictly increasing, $N(\lambda; n) \sim F_{\mathrm{KM}}(\lambda) \cdot n$ is strictly increasing. □

Proposition 4.2 (Linear growth of projectable mode count)**.** *For the LPS family at fixed p and $\lambda \leq \Lambda_0(p)$,*

$$N(\lambda;\, n) \;\sim\; F_{\mathrm{KM}}(\lambda) \cdot n \qquad \text{as } n \to \infty. \tag{25}$$

For the three ADE eigenvalue levels $\lambda_1 < \lambda_2 < \lambda_3$ of [5], define

$$c_i(p) \;:=\; F_{\mathrm{KM}}(\lambda_i) \;=\; \int_{\lambda_-}^{\lambda_i} \rho_{\mathrm{KM}}^{(p)}(x)\, \mathrm{d}x, \qquad i = 1, 2, 3, \tag{26}$$

satisfying $0 < c_1(p) < c_2(p) < c_3(p) \leq 1$. Since $\lambda_2^{\mathrm{norm}} = 1$ for all three-level ADE cases and $F_{\mathrm{KM}}(1) = 1/2$ by symmetry of the Kesten–McKay density,

$$c_2(p) \;=\; \tfrac{1}{2} \qquad \text{for all } p. \tag{27}$$

This is a structural identity, not a numerical coincidence.

4.4 Stabilisation ranks and mass formula

By Lemma 4.1, Definition 2.3 is well-posed. Inverting Proposition 4.2:

$$n_{\mathrm{proj}}(\lambda_i) \;\sim\; \frac{k\, \dim \rho_{\lambda_i}}{c_i(p)}. \tag{28}$$

The effective stabilisation mass and inter-generation mass ratio are therefore

$$\mathcal{M}_i \;\sim\; \frac{c_i(p)\, E_{\mathrm{P}}}{k\, \dim \rho_{\lambda_i}}, \qquad \frac{\mathcal{M}_i}{\mathcal{M}_j} \;=\; \frac{c_i(p)}{c_j(p)} \cdot \frac{\dim \rho_{\lambda_j}}{\dim \rho_{\lambda_i}}. \tag{29}$$

The universal constant k cancels in all ratios. The first factor c_i/c_j is determined by the spectral geometry of the LPS substrate; the second factor $\dim \rho_j / \dim \rho_i$ is determined by the representation theory of the ADE group. The two mechanisms are independent and act multiplicatively.

5 Stratigraphic Amplification: Numerical Evaluation

5.1 Numerical values of $c_i(p)$

Tables 1–3 report $c_i(p)$ for the three three-level ADE cases of [5] and all admissible p. The support condition $\lambda_i \in (\lambda_-, \lambda_+)$ requires $\lambda_+ > \lambda_3$, which limits $p \leq 13$ for $2T$ (ord-3), $p \leq 29$ for $2I$ (ord-4), and $p \leq 53$ for $2I$ (ord-5). Mass ratios are $\mathcal{M}_i/\mathcal{M}_j = (c_i/\dim \rho_i)/(c_j/\dim \rho_j)$.

Table 1: Coefficients $c_i(p) = F_{\mathrm{KM}}(\lambda_i)$ for $2T$, S = ord-3, $|S| = 8$; normalised levels (0.75, 1.00, 1.50); representation dimensions (8, 9, 6).

p	λ_-	λ_+	c_1	c_2	c_3	$\mathcal{M}_1/\mathcal{M}_2$	$\mathcal{M}_2/\mathcal{M}_3$	$\mathcal{M}_1/\mathcal{M}_3$
5	0.255	1.745	0.322	0.500	0.857	0.724	0.389	0.282
13	0.485	1.515	0.219	0.500	0.996	0.493	0.335	0.165

Table 2: Same for $2I$, S = ord-4, $|S| = 30$; normalised levels (0.80, 1.00, 1.33); dimensions (25, 76, 18).

p	λ_-	λ_+	c_1	c_2	c_3	$\mathcal{M}_1/\mathcal{M}_2$	$\mathcal{M}_2/\mathcal{M}_3$	$\mathcal{M}_1/\mathcal{M}_3$
5	0.255	1.745	0.358	0.500	0.738	2.17	0.161	0.349
13	0.485	1.515	0.273	0.500	0.867	1.66	0.137	0.227
29	0.641	1.359	0.172	0.500	0.988	1.04	0.120	0.125

5.2 Ordering of the predicted hierarchy

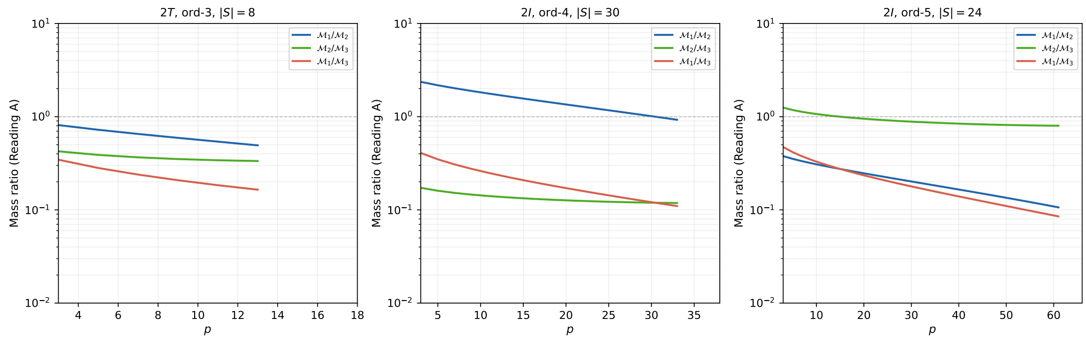

Figure 2: Predicted $\log_{10}$ mass ratios under Reading A ($\mathcal{M}_i \propto c_i/\dim\rho_i$) for $2I$ ord-4 (left) and $2I$ ord-5 (right), for all admissible primes p. Dashed horizontal lines show the observed Standard Model ratios for charged leptons, up-type quarks, and down-type quarks. The predicted ratios lie within $[-1, +1]$ on the log scale, whereas the observed ratios reach -5 for up-type quarks.

The mass ordering depends on $c_i/\dim\rho_i$. In SpectralStratigraphy [5], the admissibility envelope predicts $\mathcal{M}_1 > \mathcal{M}_2 > \mathcal{M}_3$ (lower λ = heavier). The saturation mechanism predicts the opposite for $2I$ (ord-4): the level with largest λ_3 acquires the highest $c_3 \to 1$, giving $c_3/\dim\rho_3 \gg c_1/\dim\rho_1$. This inversion holds for all admissible p and is a structural result, not a numerical accident.

5.3 Magnitude of the predicted hierarchy

All predicted ratios $\mathcal{M}_i/\mathcal{M}_j$ lie in [0.10, 2.2]. The observed ratios for Standard Model fermions range from $m_e/m_\mu \approx 1/207$ to $m_u/m_t \sim 10^{-5}$. The best LPS–KM prediction ($\mathcal{M}_1/\mathcal{M}_3 \approx 0.10$ for $2I$ ord-5, $p = 53$) falls short by approximately two orders of magnitude for leptons and four for up-type quarks.

Table 3: Same for $2I$, S = ord-5, $|S| = 24$; normalised levels (0.833, 1.00, 1.25); dimensions (54, 25, 40).

p	λ_-	λ_+	c_1	c_2	c_3	$\mathcal{M}_1/\mathcal{M}_2$	$\mathcal{M}_2/\mathcal{M}_3$	$\mathcal{M}_1/\mathcal{M}_3$
5	0.255	1.745	0.381	0.500	0.678	0.353	1.18	0.416
13	0.485	1.515	0.310	0.500	0.781	0.287	1.02	0.294
29	0.641	1.359	0.222	0.500	0.899	0.206	0.890	0.183
53	0.730	1.270	0.137	0.500	0.988	0.127	0.810	0.103

5.4 Interpretation

The LPS–Kesten–McKay saturation mechanism produces a well-defined three-level spectral stratification but fails to reproduce the observed inter-generation mass hierarchy, both in mass ordering and in magnitude.

The failure identifies two necessary extensions: a dynamical admissibility threshold and a supplementary amplification mechanism. Three open directions are described in Section 6.5.

6 Discussion

6.1 Status of the structural hypotheses

Hypothesis 2.1 identifies $c_\chi(n)$ with the isoperimetric capacity $h(G(n))$. It is the foundational physical postulate and currently lacks a derivation from the Cosmochrony axioms.

Hypothesis 2.2 ($h^2 \asymp \lambda_2$) is a property of the graph family, consistent with but not implied by the Ramanujan property. It promotes $\Lambda_{\text{proj}} \asymp h^2$ to the linear law $\Lambda_{\text{proj}} \asymp \lambda_2$.

Hypothesis 2.3 ($M(\lambda_i) = k \dim \rho_i$, k universal) is the only saturation condition internal to the framework that avoids free parameters. The constant k cancels in all mass ratios.

Of the three hypotheses, Hypothesis 2.1 is the most in need of further justification.

6.2 Role of the parameter p

The prime p controls the Kesten–McKay support width via $\lambda_\pm = 1 \pm 2\sqrt{p}/(p+1)$. In the limit $p \to \infty$, the support collapses to $\{\lambda = 1\}$, so the spectral distribution becomes increasingly concentrated around the midpoint of the spectrum. Varying p within the admissible range changes $c_i(p)$ quantitatively but not the qualitative conclusions: mass ratios remain of order one and the ordering inversion persists. The parameter p should be regarded as a structural parameter of the cascade-graph model. A natural question is whether p can be selected by the framework: the degree $d = p+1$ is related to the number of generators of $\mathrm{PSL}(2, \mathbb{F}_q)$ via the four-square decomposition of p [9]; a connection to the relational valence of the Cosmochrony substrate would constitute a selection principle.

6.3 Relation with SpectralStratigraphy

The two results of [5] used as inputs – no-go for scalar calibration and three-level ADE structure – are unaffected by the present analysis.

The present work identifies two independent components of $n_{\text{proj}}(\lambda)$: the admissibility support condition $\lambda \le \Lambda_{\text{proj}}$ and the saturation condition $N(\lambda; n) \ge k \dim \rho_\lambda$. The two mechanisms operate in different regimes. SpectralStratigraphy describes the regime in which $\Lambda_{\text{proj}}(n)$ evolves along the cascade; the present analysis shows that in the LPS model at fixed p the threshold becomes asymptotically static, leaving saturation as the only active ordering mechanism. The admissibility ordering is recovered only when $\Lambda_{\text{proj}}(n)$ varies significantly, which requires a regime where $\lambda_2(n)$ is not asymptotically constant.

6.4 Interpretation of the numerical results

The structural identity $c_2 = F_{\text{KM}}(1) = 1/2$ holds for all p and all three-level ADE cases. It follows from the symmetry $\rho_{\text{KM}}(\lambda) = \rho_{\text{KM}}(2-\lambda)$ of the Kesten–McKay density combined with the fact that the central ADE level always satisfies $\lambda_2^{\text{norm}} = 1$. This coincidence is not accidental but follows from the combinatorial normalisation of the Laplacian together with the symmetry of the Kesten–McKay density. The identity $c_2 = 1/2$ is therefore a structural alignment between the spectral geometry of the projective substrate and the representation-theoretic structure of the ADE groups.

6.5 Open directions

(O1) Dynamical threshold via joint $p(n)$, q variation.

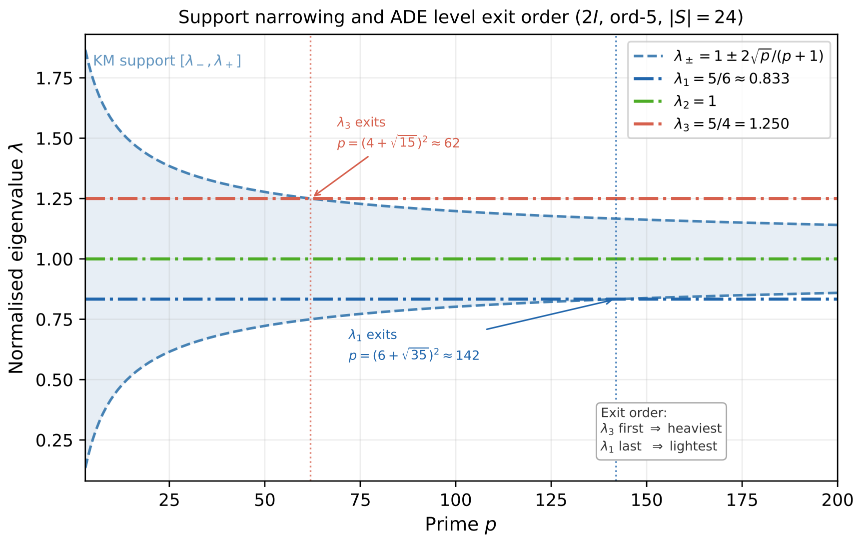

Figure 3: Kesten–McKay support edges $\lambda_\pm(p) = 1 \pm 2\sqrt{p}/(p+1)$ as functions of the prime p, with the three ADE eigenvalue levels of $2I$ (ord-5, $|S| = 24$) shown as horizontal dash-dotted lines. As p grows, the support *narrows*: $\lambda_+ - \lambda_- = 4\sqrt{p}/(p+1) \to 0$. The level $\lambda_3 = 5/4$ exits the support at $p = (4+\sqrt{15})^2 \approx 62$; $\lambda_2 = 1$ never exits (it is the midpoint of the symmetric support); $\lambda_1 = 5/6$ exits at $p = (6+\sqrt{35})^2 \approx 142$.

If p grows with n – for instance $p(n) \sim n^\beta$ for some $\beta > 0$, i.e. an effective increase of relational valence along the cascade – then $\lambda_2^*(p) = 1 - 2\sqrt{p}/(p+1) \to 1$ and the Kesten–McKay support narrows with n. Such a joint scaling would correspond physically to an increase of the effective relational valence of the substrate along the cascade. This direction is the primary candidate for addressing both the ordering inversion and the amplitude problem simultaneously.

A notable structural consequence of this mechanism concerns the ordering of stabilisation events. As $p(n)$ grows, the support $[\lambda_-, \lambda_+]$ narrows symmetrically around $\lambda = 1$. Since the ADE levels are fixed at $\lambda_1 < 1 < \lambda_3$, the level farthest from the centre exits the support first. For $2I$ (ord-5), the exit order is λ_3 (at $p \approx 62$), then λ_1 (at $p \approx 142$), while $\lambda_2 = 1$ never exits. A level that exits the support is no longer admissible and stabilises at the corresponding cascade rank; earlier exit therefore corresponds to higher stabilisation mass. The resulting mass ordering is $\mathcal{M}_3 > \mathcal{M}_1 > \mathcal{M}_2$, which for the physically relevant assignment $\lambda_3 \leftrightarrow$ first generation restores the hierarchy expected from the admissibility envelope of [5], without any additional dynamical hypothesis. This ordering result is a direct structural implication of the support narrowing and is independent of the precise growth law $p(n)$.

(O2) Hierarchical cascade. If the saturation process operates recursively across L nested levels, the per-level ratio r yields r^L. With $r \approx 0.1$ and $L = 3$, one obtains $r^3 \approx 10^{-3}$, approaching the lepton mass ratio $m_e/m_\tau \approx 3 \times 10^{-4}$.

(O3) Sub-linear energy-rank relation. Replacing $E_n = E_{\mathrm{P}}/n$ by $E_n = E_{\mathrm{P}}/n^\alpha$ with $\alpha < 1$ modifies the ratios to $(c_i \dim \rho_j)^\alpha/(c_j \dim \rho_i)^\alpha$. This reduces the amplitude problem but preserves the ordering inversion, so it does not constitute a complete solution.

7 Conclusion

This paper analysed the spectral mechanism governing the projective resolution along the projective cascade within an LPS Ramanujan graph model, and established that this mechanism, at fixed prime p, is insufficient to produce the observed mass hierarchy. The derivation of the mass hierarchy rests on the O-series transfer chain $c_\chi \to \delta_{\mathrm{pair}} \to \beta^*$ [6, 4]; the present results identify the structural obstacle and the open direction that a complete cosmological model must address. The companion paper [5] established that the representation-theoretic structure of ADE binary Cayley graphs fixes the number and order of stratigraphic levels, while the dynamical growth of $\Lambda_{\mathrm{proj}}(n)$ controls their separation. The present paper addresses this second question within the LPS model.

The main positive results are: the isoperimetric law $\Lambda_{\mathrm{proj}}(n) \asymp h(G(n))^2$ (Proposition 2.1); the Cheeger enclosure $\lambda_2^2 \lesssim \Lambda_{\mathrm{proj}} \lesssim \lambda_2$ (Corollary 2.1); the linear law $\Lambda_{\mathrm{proj}} \asymp \lambda_2$ under spectral-isoperimetric saturation (Corollary 2.2); the linear growth $N(\lambda; n) \sim F_{\mathrm{KM}}(\lambda) \cdot n$ (Proposition 4.2); and the factorised mass ratio formula

$$\frac{\mathcal{M}_i}{\mathcal{M}_j} = \frac{F_{\mathrm{KM}}(\lambda_i)}{F_{\mathrm{KM}}(\lambda_j)} \cdot \frac{\dim \rho_{\lambda_j}}{\dim \rho_{\lambda_i}},$$

in which the structural identity $F_{\mathrm{KM}}(1) = 1/2$ holds universally.

Two limitations are established: the saturation mechanism inverts the mass ordering expected from the admissibility envelope, and the predicted mass ratios lie in [0.10, 2.2], far below the observed 10^{-3}–10^{-5} range. Both limitations are traced to the asymptotic constancy of Λ_{proj} in the LPS model at fixed p.

The primary open direction is direction (O1): allowing $p(n) \to \infty$ with n, so that the Kesten–McKay support narrows progressively, the ADE levels leave the support in a rank-dependent order, and the admissibility and saturation conditions are coupled. This regime, in which the effective relational valence of the substrate increases along the cascade, is the natural candidate for recovering both the correct mass ordering and the observed amplification, and will be the subject of the next paper in the admissibility sub-programme.

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